

Research Interests	Implementation and simulation of Networked Systems and the Internet of Things.	
Education		
Ph.D.	Indian Institute of Technology Kanpur (IIT-K) Department of Computer Science & Engineering Advisor: Dr. Amitangshu Pal & Dr. Debopam Bhattacharjee – CGPA: 8.75/10	Kanpur, Uttar Pradesh 2022 – Present
M.Tech.	University of Hyderabad (UoH) School of Computer and Information Sciences collaboration with IDRBT Thesis title: An Integrated and Standards-based Fog Computing Federation Framework Advisor: Prof. Satish Narayana Srirama – CGPA: 9.61/10	Hyderabad, Telangana 2020 – 2022
B.Tech.	West Bengal University of Technology (WBUT) Computer Science & Engineering, JIS College of Engineering CGPA: 8.6/10	Kolkata, West Bengal 2014 – 2018
Research Experience	Cloud and Smart Labs Advisor: Prof. Satish N. Srirama (UoH) An Integrated and Standards-based Fog Computing Federation Framework: Dynamic deployment of services on the fog nodes out of the box on the fly with OASIS TOSCA standard. GitHub .	
Teaching Experience	Course Instructor - Advanced Python Programming 2024-25 - CSE-S208 - Data Structure Using Python 2023-24-II Teaching Assistant, Department of CSE - ESC111M/112M - Fundamentals of Computing - I & II 2024-25-II - CS724 - Sensing, Communications and Networking for Smart Wireless Devices 2023-24-I - CS610 - Programming for Performance 2022-23-I Teaching Assistant, eMasters in Financial Technology & Management - MBA977 - IT Platforms, Cloud Computing, and IoT 2023-24 Teaching Assistant, eMasters in Cyber Security - CS971 - Computer Networking II 2022-23 - CS963 - Computer Networking I 2022-23 Tutor, Department of CSE - ESC111M/112M - Fundamentals of Computing - I & II 2022-23-II	
Conference Publications	LEOCraft: Towards Designing Performant LEO Networks – S. Basak, A. Pal, D. Bhattacharjee CosmicDance: Measuring Low Earth Orbital Shifts due to Solar Radiations – S. Basak, A. Pal, D. Bhattacharjee Fog Computing out of the Box with FogDEFT Framework: A Case Study – S. N. Srirama, S. Basak	
2025 USENIX ATC		
2024 ACM IMC		
2022 IEEE Cloud		

Journal Publications

2023 ACM IJNM

Fog computing out of the box: Dynamic deployment of fog service containers with TOSCA – S. Basak, S. N. Srirama

Workshop Publications

2025 SIGCOMM LEO-NET

An investigation of Starlink's performance during the May '24 solar superstorm – S. Basak, A. Pal, D. Bhattacharjee

2023 MobiCom LEO-NET

Exploring Low-Earth Orbit Network Design – S. Basak, A. Pal, D. Bhattacharjee

Talks

2025 USENIX ATC

LEOCraft: Towards Designing Performant LEO Networks

2025 IIT Kanpur

[CS801 Research Colloquium](#) – Measuring Orbital Shifts Due to Solar Radiation

2024 ACM IMC

CosmicDance: Measuring Low Earth Orbital Shifts due to Solar Radiations

2024 IIT Kanpur

Beaming Internet connectivity directly from Space - State of the Art (SOTA) Seminar

Honors and Scholarships

Prime Minister's Research Fellowship (PMRF) – Cycle 11 2023 – 2026

AICTE scholarship in M.Tech. (Ministry of Education – India) 2020 – 2022

GATE qualified with 95%tile 2019 and 2020

Smart India Hackathon finalist (KL University, Guntur, Andhra Pradesh) 2017

Technical Skills

Programming:

Proficient in: C, Python – Familiar with: Flask, Verilog, Dart, Flutter, Solidity, JavaScript, Java

Tools & Softwares:

Gurobi Optimizer, Ansible, Docker, Docker Swarm, TOSCA, OpenStack, QEMU/KVM

Hardware:

Raspberry Pi, Arduino

Referees

IIT Kanpur

Dr. Amitangshu Pal, Assistant Professor

Department of Computer Science & Engineering

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Microsoft Research

Dr. Debopam Bhattacharjee, Senior Researcher

E-mail: debopamb@microsoft.com, **Web:** bdebopam.github.io

University of Hyderabad

Prof. Satish Narayana Srirama, Professor (Member IEEE)

School of Computer and Information Sciences

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